Finandex Whitepaper

Global Financial Market Indexing Insights — 27 data-grounded chatbots for markets, economics, and law

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1. Summary

Finandex is a progressive web app (PWA) with Android and iOS versions that offers 27 data-grounded chatbots — called "Dex" — covering global financial markets, Indonesian markets and economy, crypto and macro, startups and venture capital, cost of living, social issues, public sentiment, and legal information.

Its core principle is simple: every number comes from a verified dataset, is computed by code tested against reference (golden) values, and is always shown with a chart, its source, its as-of date, and an explanation of how it was calculated. Finandex does not invent numbers, does not give investment recommendations, and does not replace professional advice.

- 27 Dex in 8 functional categories
- 42 dataset files verified by SHA-256
- 361 read-only analytics tools shared by the chatbot and the app screens
- Answers with charts, a "Why this answer?" panel, citations, and as-of dates
- Core data works offline; Indonesian and English

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2. Problem and goals

Market and economic data is scattered across many sources with different formats, quality, and dates. General chatbots answer fluently but often cannot show where a number came from, how current it is, or how it was computed.

Finandex closes that gap: answers traceable to source files, explicit data-cleaning rules, documented formulas, and openly stated limitations (not real-time, samples, synthetic data, provider labels).

- Transparency: source, as-of date, and method are always visible.
- Consistency: chat numbers equal feature-tab numbers because both use the same tools.
- Safety: no personal data requested; seed phrases, private keys, card numbers, and national IDs are blocked.
- Accessibility: WCAG 2.2 AA, two languages, usable offline.

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3. Product: 27 Dex

Each Dex has its own menu: a Chat tab (first), feature tabs for its domain (charts, rankings, search, comparisons), and a Data tab (last) listing the exact source files, coverage, cleaning rules, quality report, and licence. The home screen shows all Dex as a feature map grouped by similar function.

Dex	Category	Status	As of	Licence
Finandex AI Global market assistant: indices, commodities, crypto, currencies (main chatbot & router)	Global Markets & Equities	live	19 October 2026	per provider
NASDAQDex 500 NASDAQ stocks, 1962–2026	Global Markets & Equities	live	18 September 2026	per provider
SnP500Dex 233 S&P 500 stocks: total return, dividends, volume	Global Markets & Equities	live	31 August 2026	per provider
CommerceDex 43 global e-commerce giants: prices, fundamentals, macro	Global Markets & Equities	live	13 May 2026	CC BY 4.0
IPODex Global IPOs 2019–2026: India, US, UK & 100+ countries	Global Markets & Equities	live	18 June 2026	CC BY-NC 4.0
IDFinanDex Indonesia multi-asset & macro: IHSG, USD/IDR, BI rate	Indonesian Markets & Economy	live	29 May 2026	per provider
RupiahDex Daily USD/IDR exchange rate since 2001	Indonesian Markets & Economy	live	20 October 2026	per provider
IDXDex Indonesia Stock Exchange equities: prices, fundamentals, IPOs	Indonesian Markets & Economy	live	17 May 2024	per provider
ReksaDex Indonesian mutual funds: AUM, investment managers, performance	Indonesian Markets & Economy	live	29 May 2026	per provider
CryptDex Crypto: prices, market snapshot, DeFi, protocol fees, bot education	Crypto & Global Macro	live	20 October 2026	mixed
CentralBankDex Central bank balance sheets & global liquidity (Fed, ECB, BoJ, 167 countries)	Crypto & Global Macro	live	16 October 2026	per provider

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Dex	Category	Status	As of	Licence
InflationDex Inflation & cost of living across countries	Crypto & Global Macro	live	2024	per provider
StartupDex Global startup funding rounds 2019–2026	Startups, VC & Companies	live	12 June 2026	CC BY-NC 4.0
VCDex Startup investor directory 2026	Startups, VC & Companies	live	2019–2026	CC BY-NC 4.0
VCNetworkDex Co-investment network among VC investors	Startups, VC & Companies	live	2019–2026	CC BY-NC 4.0
OpenCompanyDex Directory of 100k companies (Open Company Database sample)	Startups, VC & Companies	live	2026-10	CC BY-NC 4.0
SalaryDex Global salaries by skill & role (100k job-posting sample)	Cost of Living, Work & Household	live	2026-06	CC BY-NC 4.0
CostLivingDex Cost & quality of life in 148 cities (synthetic data)	Cost of Living, Work & Household	live	15 June 2024	per provider
IDHousingDex Indonesian subsidised & commercial housing (project listings)	Cost of Living, Work & Household	live	–	per provider
IDFoodDeliveryDex Price benchmark of 10k Indonesian food-delivery menu items	Cost of Living, Work & Household	live	–	per provider
PovertyDex Multidimensional Poverty Index (MPI) for 108 countries	Social & Development	live	per country	per provider
PopulationDex Global population news coverage 2019–2026	Social & Development	live	29 July 2026	per provider
HomelessnessDex Global homelessness news coverage 2019–2026	Social & Development	live	29 July 2026	per provider
HousingDex Global housing news coverage 2019–2026	Social & Development	live	23 July 2026	per provider

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Dex	Category	Status	As of	Licence
FintechDex Fintech product adoption & reviews from 14 platforms	Sentiment & Personal Finance	live	8 September 2026	per provider
PersonalFinanDex Personal finance & financial independence discussions	Sentiment & Personal Finance	live	16 September 2026	per provider
My LAIwyer Legal information: 1945 Constitution, old & new Criminal Procedure Code, Law 1/1946	Law	partial	per document	per provider

> "Partial" for My LAIwyer means some legal sources are still awaiting documents (e.g. the amended 1945 Constitution, the new Criminal Code, the Civil Code). CC BY-NC 4.0 Dex are hidden automatically in the commercial distribution.

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4. Data and pipeline

42 dataset files are verified against a manifest (byte size, SHA-256, and zip contents) before processing. Legal documents are identified by content, not file name. Third-party content inside datasets (READMEs, provider notes, review text) is treated as data, never as instructions.

The TypeScript ETL streams CSV, Parquet, and PDF inputs, applies per-Dex cleaning rules, validates every output file against a schema, limits precision to 10 significant digits, and rejects NaN/Infinity. Output is split into "core" (small, first view and offline) and "heavy" (full daily series, matrices, legal indexes).

Rule	Example
One-day round-trip spikes removed	NASDAQ, IDX, crypto prices (r > +100% then < −50%, or the reverse)
Systemic dates → reliable_from	NASDAQ tickers moving > 40% on a shared anomalous date
Iterative FX bad-print removal	USD/IDR: round-trip spikes & plateaus vs an 11-row rolling median
Holiday carry-forward excluded	IDFinanDex market series
Double counting prevented	DeFi protocol TVL is never summed as a market total
Aggregates are not countries	World Bank aggregate codes excluded from country rankings
Privacy	Authors, follower counts, mentions, and handles removed from social data

The data output is about 610 MB (core about 39 MB). Every file is recorded with its SHA-256 in the data manifest, and a small summary (as-of, quality) keeps the first screen light.

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5. Analytics methodology

Analytics are ported one-to-one from a Python reference implementation and tested against golden values with a relative tolerance of 1e-6 (absolute 1e-9 near zero). Dates are calendar dates without time-zone conversion; month/year offsets clamp to month end.

`simple_return`	Return = end price ÷ start price − 1 (start = last close on/before the start date).
`total_return`	Total return uses Adj Close (dividends reinvested).
`cagr`	CAGR = (end ÷ start)^(365.25 ÷ days) − 1, only for periods ≥ 1 year.
`vol_1y`	Volatility 1Y = sample std. dev. of daily returns (date > L − 1 year) × $\sqrt{N}$ (N = 252; crypto 365), min 60 returns.
`max_drawdown`	Drawdown = price ÷ running peak − 1; max drawdown = the minimum, with peak, trough, and recovery dates.
`ath`	ATH = highest close in the data; distance = last ÷ ATH − 1.
`calendar_return`	Calendar-year return = year-end close ÷ previous year-end close − 1; first and current years flagged partial.
`correlation`	Pearson correlation of daily returns computed after aligning prices on common dates (inner join), min 60 returns.
`beta`	Beta = cov(stock, benchmark returns) ÷ var(benchmark returns), prices aligned first, min 120 returns.
`rebase_100`	Rebase 100 = price ÷ price on the common start date × 100.
`dividend_ttm`	TTM dividends = sum of dividends dated > L − 1 year; TTM yield = TTM ÷ last Close.
`fx_lookup`	Rate = last close on/before the requested date (after removing bad prints).
`monthly_return`	Monthly return = NAV/unit ÷ previous month NAV/unit − 1 (consecutive months only).
`ratio_fundamental`	ROE = net income ÷ equity; NPM = net income ÷ revenue; DER = total liabilities ÷ equity (latest FY).
`median`	Median of the n rows matching the filter (missing values excluded).

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6. Architecture and how the chatbot works

One TypeScript codebase: ETL → validated static data → read-only tools per Dex. The same tools run on the server for the chatbot and in the browser for feature tabs, so numbers are identical in both.

- Chat requests are strictly validated (schema, ≤ 64 KB, ≤ 20 messages), rate-limited per device/IP, and processed in a loop of ≤ 8 turns and ≤ 12 tool calls.
- Responses stream (SSE): tool status, text, chart artifacts, calculation method, citations.
- AI mode: Claude chooses tools and writes the analysis from tool results. Data mode (no AI model): a deterministic router selects tools from entities, keywords, options, and countries in the question, and the answer is assembled directly from tool results.
- The "Why this answer?" panel is generated deterministically from method events — not from the model's internal reasoning.

Layer	Technology
Interface	React, Vite, Tailwind, ECharts, pdf.js
Offline & native	Service worker (core precache, heavy cache ≤ 300 MB), Capacitor Android & iOS
Server	Web Fetch API functions (Vercel/Netlify), CORS allowlist, rate limiting, security headers & CSP
Model	Claude (Anthropic) via strict-schema tool use; optional

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7. Quality and verification

Every change must pass automated quality gates. Results for the 22 October 2026 build:

Gate	Result
Lint & typecheck (strict)	0 errors, 0 warnings
Unit, golden, tools, UI components	729 tests passing; 96.4% line coverage
Golden reference values	62/62 tests, 31 golden folders
Python ↔ golden ↔ TypeScript parity	59/59 reference outputs match
Tools & tab panels	541/541
Chatbot evaluation (mock)	128/128, safety items 100%
E2E (Playwright)	123/123 (desktop & phone)
Accessibility (axe)	0 serious/critical violations; AA token contrast
Performance (Lighthouse mobile, home)	Performance 90, Accessibility 100, Best Practices 100, SEO 100
Initial bundle size	165 KB gzip (budget 250 KB)
Security & licences	npm audit 0 vulnerabilities; secret scan clean; no GPL/AGPL licences

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8. Security and privacy

- No accounts and no third-party analytics; chat history, watchlist, and preferences stay on the device and can be deleted.
- A sensitive-data detector runs on the device and on the server; messages with seed phrases, private keys, card numbers (Luhn), or national ID numbers are blocked.
- The server does not store message contents; logs contain only request ID, Dex, tool names, latency, tokens, and error codes.
- Dataset content reaches the model only as sanitised tool results; instructions inside data are ignored (prompt-injection defence).
- Secrets live only in server environments; the client bundle is scanned automatically.
- The privacy policy refers to Indonesia's Personal Data Protection Law (No. 27 of 2022); chat consent is versioned.

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9. Data licensing and distribution

Every Dex shows its dataset licence. CC BY-NC 4.0 datasets (SalaryDex, StartupDex, OpenCompanyDex, VCNetworkDex, VCDex, IPODex) are for non-commercial use only; the commercial build profile hides them automatically. Datasets without a stated licence are marked "per provider" and must be confirmed before public or commercial distribution.

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10. Limitations

- Historical data with different as-of dates per Dex; not real-time.
- Some datasets are samples (2%–38%); figures describe the sample, not the population.
- CostLivingDex uses clearly labelled synthetic data.
- News-coverage Dex (PopulationDex, HomelessnessDex, HousingDex) measure news volume, not real-world events.
- Data mode without an AI model understands common question patterns only; complex free-form questions need AI mode.
- Legal information is limited to the available documents; not legal advice.

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11. Roadmap

- Enable AI mode in production and run the live evaluation (target ≥ 95%, safety 100%).
- Add legal sources: amended 1945 Constitution, new Criminal Code (Law 1/2023), Civil Code, HIR/RBg.
- Confirm dataset licences for public/commercial distribution.
- Release the Android and iOS apps through the app stores.
- Periodic data updates with the same verification and golden tests.

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12. Closing and disclaimer

Finandex is designed and developed by Dea Saka Kurnia Putra — A BRIGHTER TOMORROW THROUGH MEANINGFUL WORK.

> Finandex presents historical data for information and education. It is not investment advice, a solicitation to buy or sell securities or crypto assets, or financial or legal advice. Past performance does not guarantee future results. Data is not real-time and may contain errors from its sources.



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